

ASIF SHAHRIAR

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RESEARCH INTERESTS

- Computer security, AI for security, Security for AI, Adversarial ML, Secured Agentic Frameworks
- Natural Language Processing, Large Language Models, Retrieval Augmented Generation, Trustworthy Generative AI

EDUCATION

Bangladesh University of Engineering & Technology (BUET)

2019 - 2024

B.Sc in Computer Science & Engineering

- CGPA: **3.85/4.0**
- CSE major-only CGPA: **3.91/4.0**
- **Dean's List** award for academic excellence

PUBLICATIONS

5GPT: 5G Vulnerability Detection by Combining Zero-Shot Capabilities of GPT-4 With Domain Aware Strategies Through Prompt Engineering

A. Shahriar, S. J. Hisham, K. M. A. Rahman, R. Islam, M. S. Hossain, R. H. Hwang, and Y. D. Lin

IEEE Transactions on Information Forensics and Security (IEEE TIFS), 2025

- We demonstrate that out-of-the-box GPT-4 is capable of identifying high-level logical inconsistencies, and ambiguous protocol rules; but suffers from hallucinations and struggles to capture deep protocol-level issues
- We introduce a novel domain-aware strategy to teach GPT-4 about security properties and hazard indicators from related works. Showed that domain-aware GPT-4 successfully identifies sophisticated multi-state and cross-procedure attacks, cryptographic and integrity violations, message spoofing, injection, privacy and identity exposure, and resource management exploits, while reducing false-positives
- Altogether we identified 47 potential vulnerabilities in 5G specifications. We tested 14 of them and confirmed 9 vulnerabilities, demonstrating the effectiveness of LLM-based automated vulnerability discovery while revealing room for improvement

Inceptive Transformers: Enhancing Contextual Representations through Multi-Scale Feature Learning Across Domains and Languages

Asif Shahriar, Rifat Shahriyar, M Saifur Rahman

Accepted for presentation in EMNLP 2025

- We develop an inception-style multi-scale feature extraction framework that sits on top of an encoder transformer model for enhancing the contextual representations
- Through a robust evaluation, we show that Inceptive Transformer improves both general-purpose (RoBERTa, DeBERTa, ModernBERT, XLM-RoBERTa) and domain-pretrained (BERTweet, BioBERT, CT-BERT, BanglaBERT) baselines by up to **14%** in FIVE different tasks (Wilcoxon signed-rank test + 10-fold cross validation for credibility)

PRE-PRINTS UNDER REVIEW

A Survey on Agentic Security: Applications, Threats and Defenses

Asif Shahriar, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez

ArXiv, 2025

- We present the first security-focused holistic survey of LLM-based agents (150+ papers), exploring how these agents are used to enhance security, their inherent vulnerabilities, and the countermeasures designed to protect them
- We also provide a detailed cross-cutting analysis that shows emerging trends in agent architecture while revealing critical research gaps in model and modality coverage

ONGOING RESEARCH WORKS

5G Vulnerability Detection using Retrieval-Augmented Generation 2024 - Ongoing

- Supervisors: **Dr. Md. Shohrab Hossain** and **Dr. Syed Rafiul Hussain (Penn State)**
- We propose a novel, fully automated end-to-end framework that utilizes a Retrieval-Augmented Generation (RAG) pipeline to ground LLM outputs in verified, domain-specific data to minimize hallucinations
- We also introduce a robust context retrieval mechanism to overcome the cross-section dependency challenges

Cross-modal Deception: There is More than what Meets the Eyes 2025 - Ongoing

- Supervisors: **Dr. Md. Shohrab Hossain** and **Dr. Rizwan Parvez (QCRI)**
- In traditional jailbreak attacks, user is the adversary while LLM is the victim. We aim to introduce a novel class of attacks that deceive both the user and the LLM
- The model is compromised by a hidden instruction, while the human user, who may be interacting with the model through a completely benign-looking image, is an unwitting participant in the attack

Repository-Level Vulnerability Auditing using LLM Agents 2025 - Ongoing

- Supervisors: **Dr. Rizwan Parvez (QCRI)**
- Benchmarks like SWE-bench primarily test an agent's ability to correct an existing, well-documented issue within a codebase (reactive bug-fixing)
- We investigate the more critical and complex task of **proactive security auditing**: the ability to review incoming code changes (e.g., a pull request) and identify subtle, hidden security vulnerabilities before they are merged into the main branch. This is fundamentally different and challenging because not only is it inherently ambiguous, but also requires long-context and compositional reasoning capabilities

TEACHING EXPERIENCE

Department of CSE, BRAC University 2024 – Current

Full-time Lecturer

[faculty-webpage](#)

- Artificial Intelligence (Theory and Sessional)
- Data Communications (Theory)
- Data Structures (Sessional)
- Algorithms (Sessional)

WORK EXPERIENCE

North-West Power Generation Company Limited May 2023 – June 2023

Machine Learning Internship

[github-link](#)

- Some of the ultrasonic flow-meters used by the company had engineering problems. I developed a machine learning algorithm that can predict the health status of a flow-meter given various readings, which achieved **83%** accuracy.

TOP PROJECTS

WhiskerDocs: A health-care solution for your pets June 2023 - September 2023

Software development project

[back-end](#) | [front-end](#)

- Analyze symptoms | Book a vet based on location, experience and rating | Dashboard | History recorded | Blogs | Subscription | Payment module | Appointment scheduling | Prescription template | Auto-completion
- Tools:** GraphQL | Next.js | nest.js | openstreetmap | PostgreSQL | Docker | Elasticsearch | Stripe
- project-presentation** | **partial-demo**

Machine Learning from Scratch December 2023 - February 2024

Supervised learning, unsupervised learning, deep learning

[github-link](#)

- Classification using Ensemble Learning.** Accuracy: **80%-91%** on three different datasets.
- Handwritten Letters Identification** using **Feed-forward Neural Network.** Accuracy: **91%**.
- Unsupervised clustering** using EM algorithm. PCA for dimensionality reduction.

Rasterization and Ray-tracing July 2023 - September 2023

Computer graphics project

[github-link](#)

- Designed a fully controllable magic cube that can smoothly transition between a sphere and an octahedron (a fair bit of geometry was involved)
- Developed the raster-based graphics pipeline used in OpenGL through modeling transformation, view transformation, projection transformation, and clipping & scan conversion using Z-buffer algorithm
- Implemented a ray-casting and recursive ray-tracing application using OpenGL and Phong lighting model for rendering a photorealistic 3D world preview with various geometric objects under appropriate illumination

DC-Vegas: A Congestion Control Algorithm for Datacenters

March 2023

Networking project with NS2 simulation

[github-link](#)

- Aims to achieve the excellent performances of DC-TCP congestion control algorithms in datacenters at the low deployment cost of TCP-Vegas algorithm
- NS2 simulation shows that DC-Vegas can produce better throughput and packet delivery ratios than TCP-Vegas

Subset C Compiler

June 2022 - August 2022

Compiler project

[github-link](#)

- Given a .c input file, this compiler scans the .c file for specific tokens, performs a syntax and semantics analysis, and finally, if the source code does not contain any error, generates assembly code for Intel 8086 assembly language
- **Tools:** C++ | Lex | Yacc | Emu8086

Fingerprint-based Automated Attendance System

August 2022

Microcontroller project

- Fingerprint based attendance system with a real-time clock module for cut-off entry time. Stores attendance in a text file.
- **Hardware:** Arduino Uno | R305 Fingerprint sensor | RTC Module DS3231 | SD card Module | LCD Display

TECHNICAL SKILLS

Languages: Python • Java • Javascript • C • C++ • ~~W~~TeX • Assembly

Development: Next.js • nest.js • Node.js • React • GraphQL • Docker • HTML • CSS • Bootstrap

Database: OracleDB • PostgreSQL

Machine-Learning: NumPy • Pandas • Scikit-learn • SciPy • Matplotlib • Seaborn

Deep-Learning: PyTorch • TensorFlow • Convolutional Neural Networks (ResNet, InceptionNet, ViT) • Recurrent Neural Networks (RNN, LSTM, GRU) • Transformers • Language models • LLMs • RAG

Networking & Security: 5G • LTE • Cryptography • EDR • NS2 • Open5GS • UERANSIM • 5GReplay • Scapy • CPPCheck • WireShark

HONOURS AND AWARDS

- **Best paper award** in 11th International Conference on Networking, Systems, and Security (NSysS 2024)
- **Dean's List award** for academic excellence in undergrad
- **Talent-pool scholarship** in Higher Secondary (**10th** in Dhaka Board)
- **Talent-pool scholarship** in Secondary (**26th** in Dhaka Board)

REFERENCES

- **Dr. Md. Shohrab Hossain**, *thesis supervisor, co-author*
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- **Dr. M Saifur Rahman**, *research supervisor, co-author*
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- **Dr. Ying-Dar Lin**, *research supervisor, co-author*
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- **Dr. Md. Rizwan Parvez**, *research supervisor, co-author*
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